

Additional Exercises 1

Limits & Continuity of Multivariable Functions

Mirrors: List 2 (limits) & List 3 (continuity) | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Find and sketch the domains of the following functions

- (a) ★ $f(x, y) = \ln(x^2 - y)$
- (b) ★ $f(x, y) = \sqrt{9 - x^2 - y^2}$
- (c) ★★ $f(x, y) = \arcsin\left(\frac{x + y}{x - y}\right)$
- (d) ★★ $f(x, y) = \ln\left(\frac{x^2 + y^2}{x^2 - y^2}\right)$
- (e) ★★★ $f(x, y) = \sqrt{\ln(x^2 + y^2 - 1)}$

Exercise 2: Find and sketch at least 4 level curves of each function

- (a) ★ $f(x, y) = x^2 - y^2$
- (b) ★ $f(x, y) = \frac{y}{x}$
- (c) ★★ $f(x, y) = x^2 + 4y^2$
- (d) ★★ $f(x, y) = \ln(x^2 + y^2)$

Exercise 3: Find the limits or show that they do not exist

- (a) ★ $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2}$
- (b) ★ $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2 + 3y^4}{x^2 + y^2}$
- (c) ★ $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \cos\left(\frac{1}{xy}\right)$

- (d) ★★ $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^4 + y^2}$
- (e) ★★ $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1}$
- (f) ★★ $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^4)}{x^2 + y^2}$
- (g) ★★ $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow 3}} \left(1 + \frac{1}{x}\right)^{x+y}$
- (h) ★★★ $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{xy} \cdot \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right)$
- (i) ★★★ $\lim_{(x,y) \rightarrow (0,0)} \frac{x^5 - x^2 y^3}{x^4 + y^4}$
- (j) ★★★ $\lim_{(x,y) \rightarrow (0,0)} \frac{x^5 - \cos(x^2 + y^2) + 1}{(x^2 + y^2)^2}$

Exercise 4: Find the maximal domain on which each function is continuous

- (a) ★ $f(x, y) = \begin{cases} \frac{x^2 + 2y^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 1, & (x, y) = (0, 0). \end{cases}$
- (b) ★ $f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$
- (c) ★★ $f(x, y) = \begin{cases} \frac{\sqrt[3]{y} \sin(x^2)}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$
- (d) ★★ $f(x, y) = \begin{cases} (x + y) \sin \frac{1}{x^2} \cos \frac{1}{y}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$
- (e) ★★ $f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$
- (f) ★★ $f(x, y) = \begin{cases} \frac{x^4 y^2}{x^4 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$

$$(g) \quad \star\star\star\star \quad f(x, y) = \begin{cases} \frac{x^4 - x^3y - xy^3 + y^4}{\sqrt{x^6 + y^6}}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

$$(h) \quad \star\star\star\star \quad f(x, y) = \begin{cases} (x^2 + y^2) \sin \frac{1}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 1, & (x, y) = (0, 0). \end{cases}$$

★ *Polar or direct substitution*
 ★★ *Squeeze, asymptotic equivalence, or two-path test*
 ★★★ *Creative path or algebraic manipulation required.*